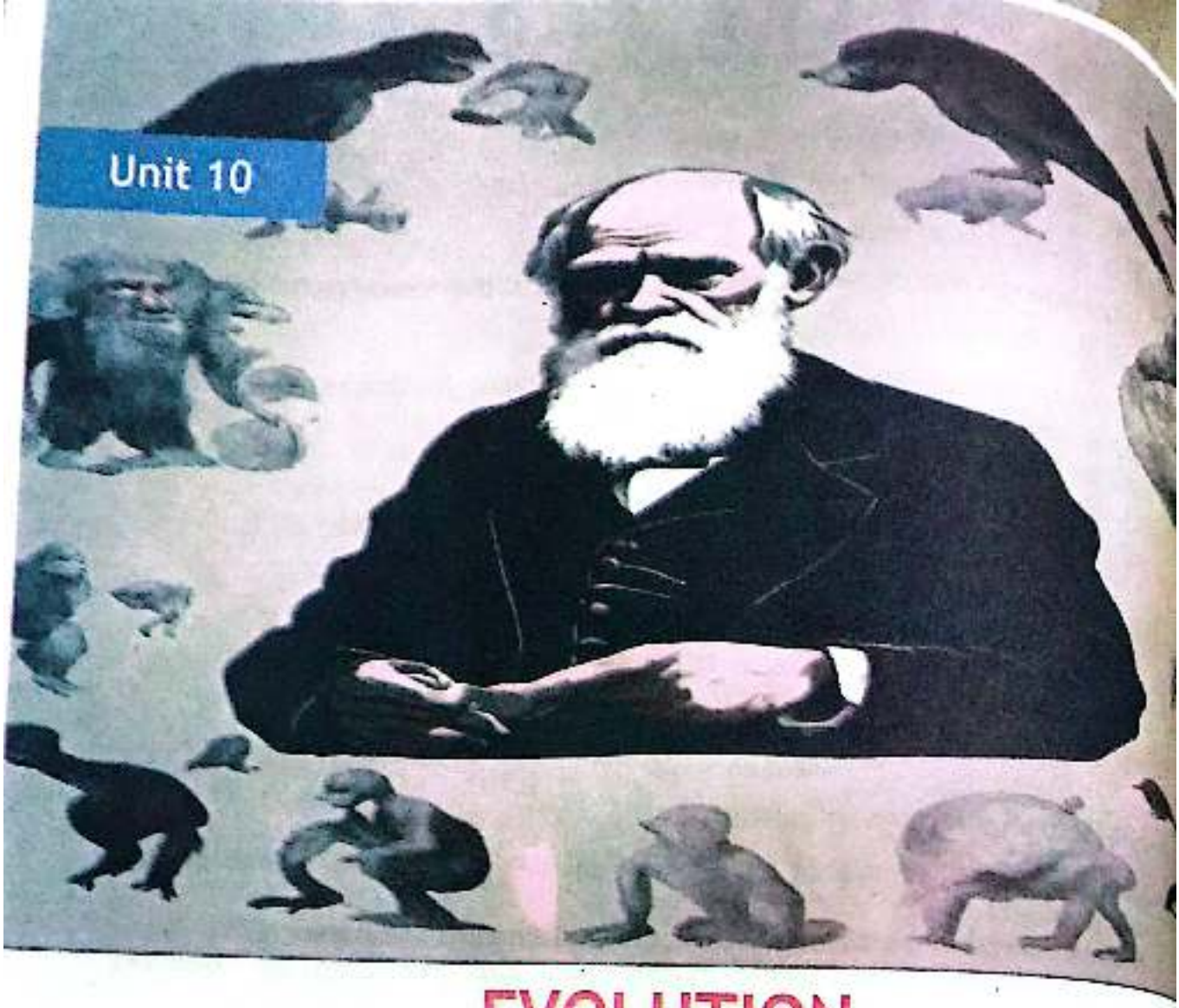


Unit 10

EVOLUTION

SLOs: After completing this lesson, the student will be able to:

1. Explain the theory of evolution by natural selection with examples
2. Discuss briefly the observations Darwin made during his voyage on HMS Beagle
3. Describe the evidence of evolution with regards to the following:
   - Palaeontology (fossil record)
   - Comparative anatomy (homologous structures, vestigial structures)
   - Selective breeding
4. Describe speciation
5. Describe sources of variations which can lead to speciation and evolution

CamScanner

The number of species of living things on earth currently exceeds a million. Each specie has its own manner of living and working. Some species appear to be highly similar to one another, whereas others do not. In many ways, a horse and a zebra are similar. Both of them, however, have some traits not present in the other. Through the process of progressive transformation, all of the species alive today evolved from earlier ones. One definition of evolution is a change that occurs to living things over time.

10.1 VARIATION AND EVOLUTION

10.1.1. VARIATION

The term variation refers to the differences in traits displayed by members of the same species. For instance, although all domestic cats are members of the same species, they differ greatly in terms of size, coat colour, eye colour, hair length, and other characteristics. Variations are beneficial for living organism as it helps in increasing their chances of survival of the particular organisms. These are desirable variations as it helps them be more advanced and immune. Variations can sometime lead to a genetic defect hence becoming a harmful and an undesirable variation.

Figure10.1: Variations in domestic cats

You are already aware of the meiotic process. Meiosis along with other factors serve as the main sources of variation. They are as follows:

1. Genetic recombination results from crossing over during gamete development.
2. Different chromosome combinations in gametes are formed by independent collections of chromosomes.
3. The process of fertilization in which one of the countless sperms fuses with an egg.
4. A mutation in the gamete-forming germ cells.
5. Gene flow, or the transfer of genes from one population of a species to another.

10.1.2. EVOLUTION

According to scientists, the Earth is more than four billion years old. New life forms developed from unicellular organisms. All current living things developed gradually from earlier ones. Evolution is the gradual alteration of a population's or species' traits over time in response to the environmental conditions. Evolutionary changes can always be passed on to next generations.

CamScanner

How variation promote evolution?

Some individuals of a population may have an advantage over other members as a result of population variations. These individuals will be able to have more progeny. Consequently, the population will gradually see more instances of that specific variant. With these variations, they will have a better chance of surviving and procreating. That's evolution.

An inherited trait or group of traits that increases an organism's chances of survival in specific environmental conditions is known as an adaptation. Members of a specie's population will compete with one another for resources. Only the healthiest, fittest, and strongest will survive and reproduce. If food or water is in little supply, there is little room for sharing, and there aren't many available partners.

10.1.3. Theory Natural Selection

Natural selection is the process through which organisms who possess traits that are helpful for reproduction in a certain environment produce more children in the following generation. As a result, they gradually expand the population's genetic diversity. Natural selection theory is based on:

1. **Overproduction:** All living creatures are very fertile so that their populations could grow quickly to produce large number of offspring.
2. **Competition:** We can observe that population sizes do not rise to a very high level even after overproduction. The numbers largely stay constant. This is because food supplies and other resources are scarce. As a result, there is competition for resources and food.
3. **Variations:** No two organisms are the same. They will not have the same opportunities to survive and reproduce as others. A large portion of this diversity is heritable. Such qualities are inherited from parents through the genes of the offspring.
4. **Natural selection:** Only those organisms who live and reproduce can give rise to the next generation. And only those organisms with the desirable variations will live till reproductive age. The population will have more advantageous genetic variations and fewer harmful ones after many generations. Natural selection is essentially an elimination process. Only fittest organisms will survive.

After a five-year voyage around the world in the ship HMS Beagle, **Charles Darwin** put forth the Theory of Natural Selection in 1838. In 1859, he released a book titled "**On the Origin of Species by Natural Selection**" recording his observations.

Charles Darwin was able to create his theory of natural selection when he visited the Galapagos Islands. He found a number of finch species that varied from island to island. He noticed that the form of the birds' beaks varied depending on the kind of food they ate. Darwin believed that the processes of speciation and adaptation were linked. The overall accumulation of adaptations in different environments, isolated from the original environment by geographic barriers, would lead to the formation of a new species from an ancestral form. The two groups could differ sufficiently over many generations to be classified as two different species. Darwin identified three basic patterns in biological diversity: (1) global, (2) local, and (3) variation over time in species.

CamScanner

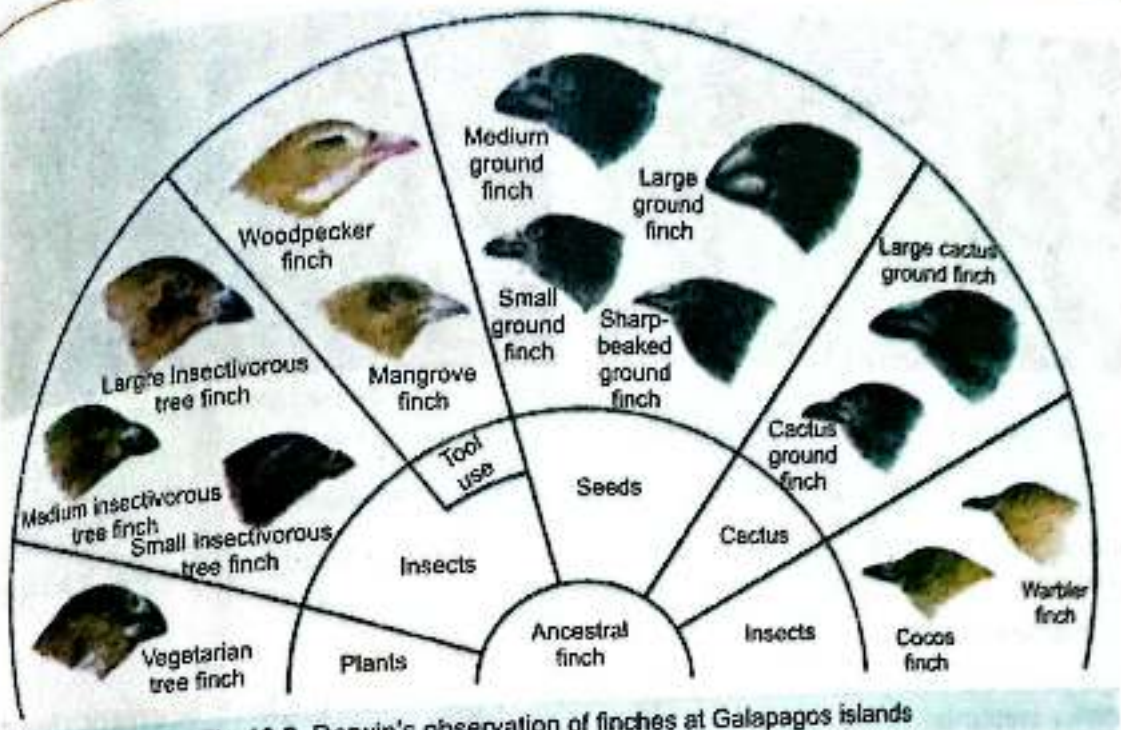


Fig 10.2. Darwin's observation of finches at Galapagos islands

Natural selection causes population-wide changes in ge ne frequencies over time; as a result, individuals with more advantageous traits will be more prevalent in the population and individuals with less advantageous traits will be less prevalent. In Britain, the Industrial Melanism of the peppered moth (*Biston betularia*) is a strong indicator of natural selection.

Analysing a case study of Variation and Selection

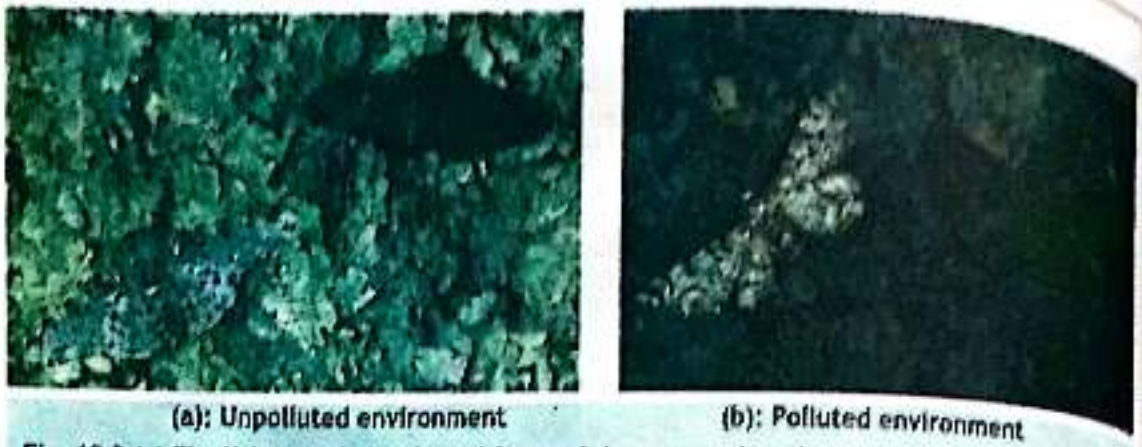
The phrase "Changes in the environment can bring about changes in a population by natural Selection" is supported by evidence.

All reported variants of the peppered moth (*Biston betularia*) in the UK up until 1848 had light colours with dark patches. A dark-coloured moth was discovered in 1848, and by 1895, dark-coloured moths made up 95% of the peppered moth population. This variety's dark colour resulted from a mutation.

The moths fly at night and rest on tree trunks throughout the day. The light-coloured lichens growing on the tree trunks blended in with the colour of the light-coloured moth, giving it excellent concealment from predator birds. As the industrial revolution progressed, sulphur dioxide pollution from coal burning drove out lichens that were growing on trees in industrial locations, exposing the darker bark.

The predator birds can now easily locate and consume the light-coloured moths that are resting on dark-coloured trunks. Light-coloured moths are "seen and eaten by the birds" when they settle on dark-coloured trunks. The birds are serving as snoopers. The dark colour was the adaptive variable that resulted in the selection advantage. 90% of the peppered moth population in England during the industrial era of pollution was black. 'Industrial Melanism' is the name given to the specific phenomenon. After the 1950s, England's industrial pollution was reduced, and as a result, the number of light-coloured moths also increased.

CamScanner



(a): Unpolluted environment (b): Polluted environment

Fig. 10.3 (a) The light and dark coloured forms of the peppered moth are resting on a lichen-covered tree. (b) The light and dark coloured forms of the peppered moth are resting on a lichen free tree trunk, which was darkened by industrial air pollution.

Analysis: (1) The variation (change in bark colour) is an illustration of variation that is subject to selection (by the birds) in a changing environment.

(2) It demonstrates the truth of evolution.

(3) It points out that evolution and natural selection operate on the basis of genetic change within a population

10.2. EVIDENCES OF EVOLUTION

The evidence for evolution is compelling and extensive. Looking at every level of organization in living systems, biologists see the signature of past and present evolution.

10.2.1. Evidence from Palaeontology

In addition to offering strong evidence that ancient species were distinct from modern ones, fossils also show the evolution of organisms across time. To determine how long ago an organism lived in relation to another, scientists classify and age fossils from all around the world. The resulting fossil record shows the evolution of form over millions of years and tells the tale of the past. For example, researchers have found extremely comprehensive records demonstrating how humans and horses have evolved.

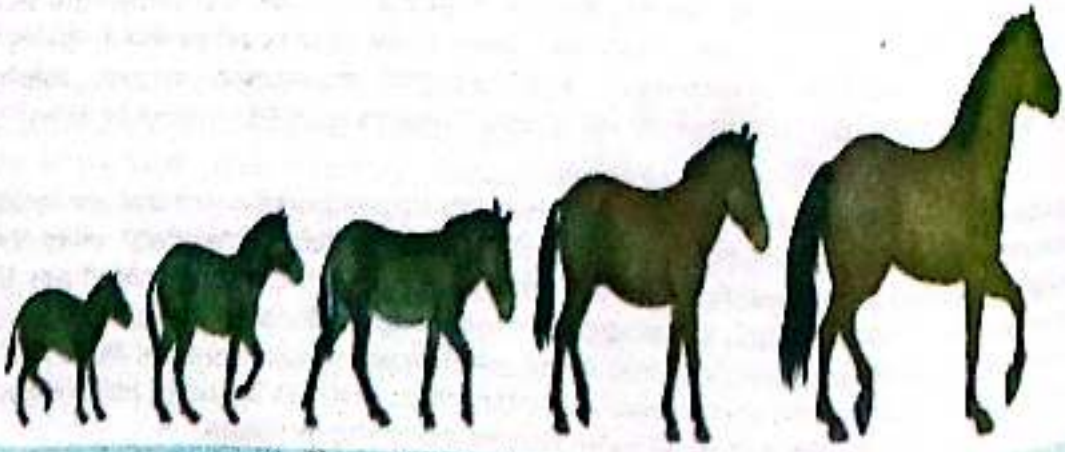


Figure 10.4: Evolution of modern horse

CamScanner

10.2.2. Evidence from Comparative Anatomy

The existence of structures in creatures with the same fundamental shape is another type of evidence supporting evolution. For example, due to their origin in the appendages of a common ancestor, the bones in the limbs of a human, dog, bird, and whale all share the same general structure. The shapes and sizes of these bones have evolved over time in different animals, but their general structure has remained the same. These similar parts are known as homologous structures.

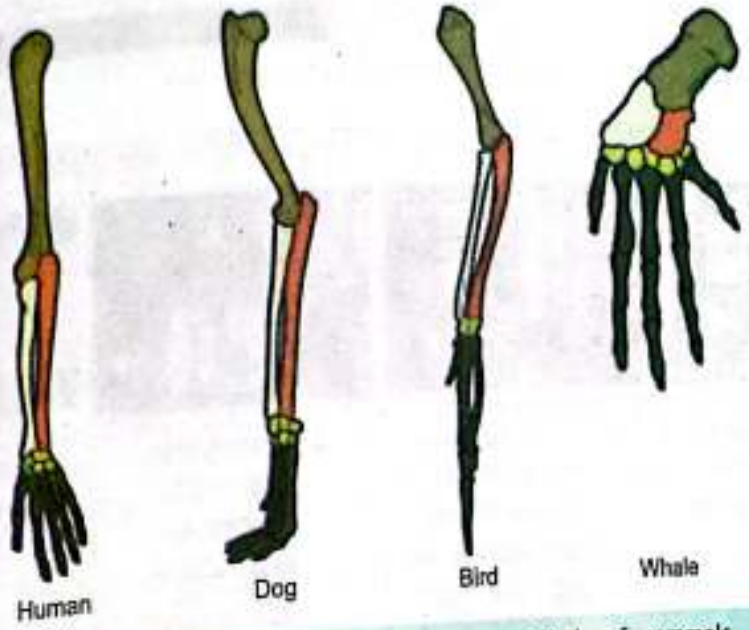


Fig 10.5: Homologous structure of bones of limbs of mammals

Some organisms possess such organs that don't appear to have any function at all now and are likely just the remains from a previous common ancestor. Vestigial organs are these inactive, non-functional structures that have lost their use over time. Wings on flightless birds, leaves on some cacti, and appendix in humans are a few examples of vestigial organs

10.3.3. Evidence from Selective breeding

Artificial selection is the practice of breeding domesticated and genetically enhanced plants and animals to produce offspring with desired genetic features. Selective breeding is the current name for artificial selection. Selective breeding and hybridization are the two techniques used to develop desired features in plants and animals.

10.2.4. Improving plants by selection

Farmers have been using selection to create plants with desired features for ages. They frequently choose varieties that will cause the plants to grow larger, more quickl y, be drought-resistant, etc.

CamScanner

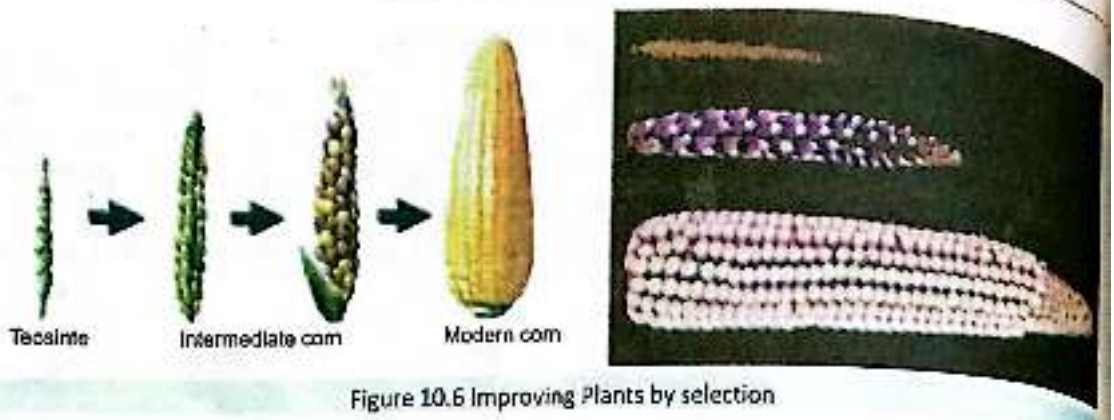


Figure 10.6 Improving Plants by selection

Artificial selection produces crops with higher yields

1. Natural variation occurs in the wild population.	2. Seeds for the next generation are chosen only from individuals with the most desirable traits.	3. Repeat this process for several generations.	4. Over time, the yield of the crop increases.

Improving animals by selection

Animals have also been the subjected to selection. Cows that generate a lot of milk and high-quality meat are chosen for selective breeding. They serve as parents for the following generation. In this manner, a breed of cow that can produce more milk than the original breed is created.

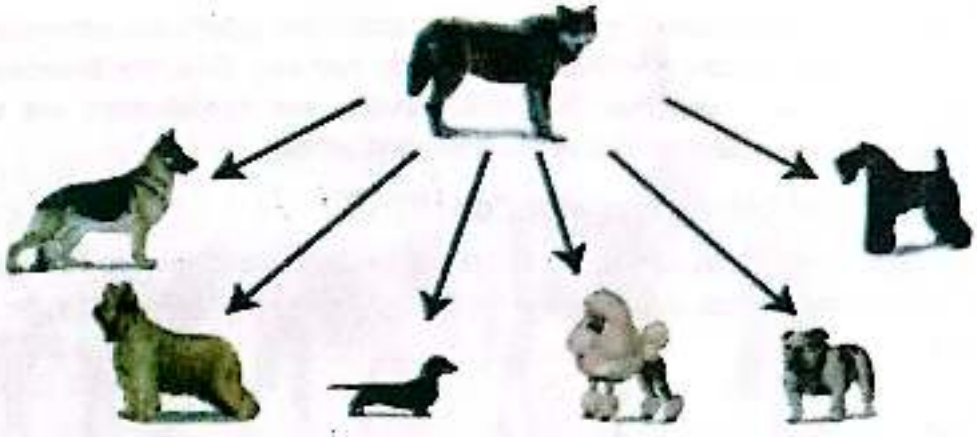


Fig. 10.7 : Dog breeds produced from wolf by artificial selection

CamScanner

Table 10.1. Comparing Natural Selection and Artificial Selection

Natural Selection	Artificial selection
Selection occurs when natural environmental condition changes.	It is selected by man.
Varieties are produced by mutations.	Varieties are produced by selective breeding.
It operates in natural population.	It operates in domesticated population.
It is a slow process and takes thousand to millions of years.	It is a fast process and results are immediate (in the next generation).
It is responsible for the great biological diversity.	It is responsible for new varieties different from starting generation.

10.3 SPECIATION

The evolutionary process by which new biological species arise is called **speciation**. Two groups that originate from the same ancestor can become very different over time if they live in different places. When a species gets split into two geographical regions, the process of speciation starts. Each adapts to its own environmental conditions. After a while, individuals from one group can no longer reproduce with the other group. Ultimately, two new species have evolved from one.

Fig 10.8. Speciation

SUMMARY

1. The differences in traits displayed by members of the same species is called variation.
2. Meiosis along with other factors serve as the main sources of variation.
3. Evolution is the gradual change of a population's or species' traits over time in response to the environmental conditions.
4. An inherited trait or group of traits that increases an organism's chances of survival in specific environmental conditions is known as an adaptation.
5. Natural selection is the process through which organisms who possess traits that are helpful for reproduction in a certain environment produce more children in the following generation.
6. After a five-year voyage around the world in the ship HMS Beagle, Charles Darwin put forth the Theory of Natural Selection.
7. The evidence of evolution is provided by palaeontology (fossil record), comparative anatomy (homologous structures, vestigial structures) and selective breeding.
8. The evolutionary process by which new biological species arise is called speciation.

EXERCISE

Section I: Multiple Choice Questions

Select the correct answer:

1. The process of ------ and------- generate variation, and -------- produces adaptation to the environment.
   A) sexual recombination ----- natural selection -----mutation
   B) genetic drift----mutation ---- sexual recombination
   C) mutation ------- sexual recombination ----natural election
   D) mutation------ natural selection-----genetic drift

2. Natural selection is sometimes described as "survival of the fittest." Which of the following most accurately measures an organism's fitness?
   A) its mutation rate
   B) how many fertile offspring it produces
   C) its ability to withstand environmental extremes
   D) how much food it is able to make or obtain.

3. The smallest biological unit that can evolve over time is
   (A) a specie
   (B) an individual organism
   (C) an ecosystem
   (D) a population

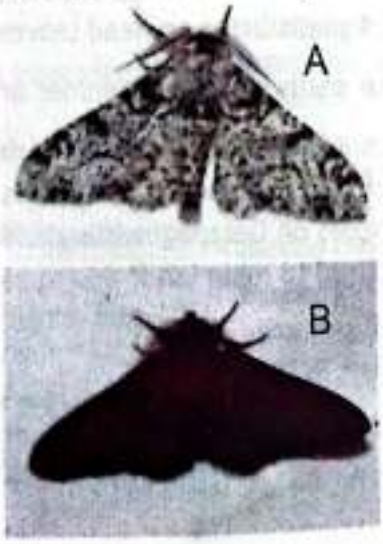
4. Which of the following pairs of structures is least likely to represent homology?
   (A) the wings of a bat and the forelimbs of a human
   (B) the haemoglobin of a baboon and that of a gorilla
   (C) the brain of a cat and that of a dog
   (D) the wings of a bird and those of an insect

CamScanner

5. Organisms produce many more offspring that can possibly survive on the limited amount of resources available to them. The offspring that are most likely to survive are those that:
   (A) are born first and grow fastest
   (B) are largest and most aggressive
   (C) have no natural predators
   (D) are best adapted to the environment
6. Natural selection operates to produce changes in:
   (A) individuals (B) populations
   (C) races (D) genes
7. Speciation is the evolutionary process by which:
   (A) a new population is formed (B) a new breed is formed
   (C) a new specie is formed (D) a new race is formed
8. A mutation is:
   (A) a change in appearance (B) a change in behaviour
   (C) a change in gene (D) a change in the environment
9. farmers have bred Neli- Ravi buffalo to produce more milk than older breeds. Which process was used to produce these cattle?
   (A) adaptation (B) natural selection
   (C) genetic engineering (D) selective breeding
10. Which results from the process of natural selection and adaptation?
   (A) artificial selection (B) evolution
   (C) reproduction (D) selective breeding

Section II: Short Answer Questions

1. Explain the homologous structures in plants.
2. Why the theory of evolution was attributed to Darwin?
3. What is meant by descent with modification?
4. The diagram shows two varieties of moths in England.



CamScanner

a. Which variety was more common when there was more air pollution? Give reason as well.

b. Which variety is more common in England nowadays? Give reason as well.

5. Define/Describe/Explain briefly:

   evolution, homologous organs, speciation, natural selection, artificial selection, industrial melanism, palaeontology, comparative anatomy, vestigial organs, selective breeding,

6. Differentiate between natural and artificial selection.

Section III: Extensive Answer Questions

1. What is variation? Explain the sources of variation.
2. Assess selection as a possible means of evolution.
3. Explain the theory of evolution by natural selection with examples.
4. Discuss briefly the observations Darwin made during his voyage on HMS Beagle.
5. Describe speciation.
6. Describe the evidence of evolution according to palaeontology.
7. How comparative anatomy proves evolution?
8. Describe the evidence of evolution according to Selective breeding.
9. A specie o snail may have a yellow, pink or brown shell. Each colour shell may have dark bands, or have no bands. The snails are eaten by birds. Following observations were made:
   - Most snails living on a plain background have no bands.
   - Most snails living on green background are yellow.
   - Most snails living on rough background have bands.

     (a) Suggest an explanation or these observations.

     (b) Predict the colour of snails living on dead leaves.

     (c) Suggest which colour snails will be in summer and autumn.
10. Many small volcanic islands are located near South America; these are called Galapagos islands. There is only one species of finches on South America continent but there are 14 species of finches on Galapagos islands. Why there are more species on Galapagos islands as compared to main land?

CamScanner